# Ultra-Efficient Image Compression via Adaptive Energy Allocation (AI/DS Track)


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## Abstract

Compression of Synthetic Aperture Radar (SAR) imagery presents several challenges compared to optical images, including high occurrence of speckle noise and the need to preserve both Amplitude and Phase components with high fidelity. Existing learned image compression methods achieve strong rate-distortion performance on SAR, but they are often impractical for resource-constrained systems or low latency requirements. In contrast, traditional codecs trade compression performance for efficiency gains. We present PACT, a lightweight neural compression architecture that incorporates Discrete Cosine Transform (DCT) preprocessing, an energy-guided hyperprior transform, and an adaptive convolutional layer for intelligent filter allocation. When applied to SAR imagery, PACT significantly outperforms VVC and competes with ELIC-DCT, a top-performing learned SAR compression model. This is achieved with 96% fewer operations and 88% fewer parameters than ELIC-DCT, demonstrating that SAR-optimized learned compression can achieve state-of-the-art quality without sacrificing real-world practicality.


## Introduction

Synthetic Aperture Radar (SAR) is an active sensor capable of producing high-resolution imagery independent of lighting conditions and cloud cover. SAR systems are widely deployed in remote sensing applications like terrain mapping, environmental tracking, agriculture, and disaster response. Unlike optical sensors, SAR captures scenes as a complex-valued signal - each pixel carries both in-phase (I) and quadrature (Q) channels, from which amplitude and phase information are derived Fig. 1. This complex representation carries physically meaningful properties that must be preserved under compression.

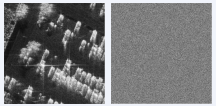

**Fig. 1.** Visualization of the amplitude and phase components of a SAR image.

The data volumes produced by modern airborne SAR platforms are substantial, requiring effective compression for storage and transmission. Standard image and video codecs like Versatile Video Coding (VVC) [1] are designed for natural images and often degrade the properties of SAR imagery that make it useful for downstream analysis. Learned image compression (LIC) has emerged as a powerful alternative, with modern neural architectures consistently outperforming classical codecs on natural image benchmarks. However, learned codecs carry significant computational overhead, which limits their practicality in resource-constrained systems like satellite or aerial SAR platforms.

Recent work has advanced LIC performance for SAR data. ELIC [2] introduced a powerful context modeling framework to achieve strong performance on natural images and establish a widely-used baseline for efficient learned codecs. ELIC-DCT [3] extended this architecture for SAR compression by incorporating DCT preprocessing, exploiting the frequency-domain structure of SAR imagery to improve on both classical codecs and general-purpose learned methods. The Asymmetric Hyperprior Transform (AHT) [4] approach introduced an energy allocation scheme to assign deeper branches of the network for higher-energy latent channels, reducing the computation needed for similar compression performance. Our Partial convolution-based Adaptive Compression in the Token domain (PACT) architecture builds on these foundations, integrating DCT preprocessing with an energy-guided transform network and AHT-style hyperprior.

Image compression performance is typically evaluated via the Rate-Distortion (R-D) tradeoff, the fundamental tension between bitrate and reconstruction quality. This tradeoff is visualized as a curve across multiple operating points, where higher bitrates correspond to higher fidelity. The Bjontegaard Delta Rate (BD-Rate) is a standard summary metric, measuring the average bitrate savings of one method over a baseline method at equivalent quality. The complexity of a neural network architecture is typically measured in both trainable parameters and multiply-accumulate operations (MACs) per pixel.

## Methods

**LIC Framework:** Modern learned image compression follows the variational autoencoder framework introduced by [5], in which an encoder network maps an input image to a quantized latent representation, and a decoder network reconstructs the image from that representation. A hyperprior network operates in parallel, encoding side information about the latent distribution that enables accurate entropy coding. The rate-distortion objective jointly minimizes image distortion and the estimated bitrate of the compressed signal. (Fig. 2)

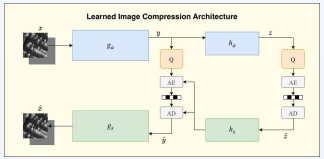


**Fig. 2.** High-level architecture of modern learned image compression systems. *Q* represents the quantization scheme, while *AE* and *AD* refer to the arithmetic encoder and decoder, respectively.

**DCT Preprocessing:** Prior to encoding, input SAR imagery undergoes a 2-D 4x4 discrete cosine transform (DCT) on each channel (I and Q). This converts the spatial-domain signal into a frequency-domain representation with a predictable energy structure, which can be exploited to intelligently allocate more complexity to higher-energy channels.

**Energy-Guided Grouped Convolution:** The model's first encoding layer sorts the 32 DCT channels by energy and partitions them into eight groups, then allocates the number of convolution filters/output channels according to each group's proportion of the total energy. For example, with 80 total output channels, a group representing 20% of the total energy is allocated 16 filters. This is implemented via grouped convolution, wherein each group is assigned an independent convolutional layer, and their outputs are concatenated to form the next layer's input. (Fig. 3)

**Asymmetric Hyperprior Transform:** The hyperprior follows an asymmetric design proposed by He et al. (2026), in which higher-energy latent channels are routed through deeper branches of the network, allocating model capacity where the signal information is most concentrated. First, the latent channels are split into four groups. During training, an Energy Allocation (EA) Loss is applied to penalize groups according to their signal energy, encouraging the groups to self-organize with decreasing energy from group 1 to group 4. Finally, the channel groups pass through network branches of different depths to de-prioritize low-energy latent channels and save redundant operations. The hyperprior outputs per-element means and scales parameterizing Gaussian distributions, which are consumed by the arithmetic coder. (Fig. 4)

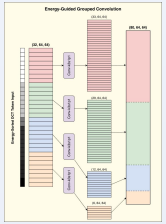


**Fig. 3.** Design for adaptive grouped convolution layer, allocating filters based on input energy.

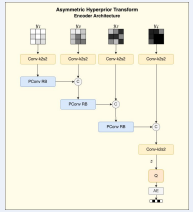


**Fig. 4.** Design for hyperprior encoder. C refers to simple concatenation.

## Methods, cont.

**Loss Function:** The rate-distortion loss is a weighted combination of bitrate and distortion terms controlled by a Lagrange multiplier $\lambda$. For SAR imagery, we measure distortion jointly by L1-SSIM Loss in the I/Q domain and SQNR Loss in the amplitude domain, controlled by their own weight parameter $\alpha$. PACT also jointly optimizes EA Loss to enable the AHT, resulting in the following final loss function:

$$\boldsymbol{L} = R + \lambda(\alpha D_{IQ} + (1\text{-}\alpha)D_{Amp}) + \gamma EA$$

**Training Details:** PACT was trained on 10,000 256x256 patches from the National Geospatial-Intelligence Agency (NGA) SAR dataset [6], using the PyTorch framework on an NVIDIA L40S GPU. Multiple operating points were obtained by training separate models at different $\lambda$ values. Models were trained in two stages: Stage 1 consisted of 250 epochs using only I/Q Loss in the distortion calculation, and Stage 2 consisted of an additional 100 epochs using both I/Q Loss and Amplitude Loss.

## Results

PACT was evaluated against several benchmarks on a representative 1024x1024 SAR patch from the NGA dataset using two primary quality metrics:
Signal-to-Quantization-Noise Ratio (SQNR) for Amplitude and Mean Absolute Error (MAE) for Phase. Bitrate is typically reported as bits per pixel (BPP), which normalizes the compressed file size according to the spatial size of the image. We choose to further normalize this value by the channel dimension, giving bits per pixel per band, but we refer to this quantity as simply BPP. Complexity is reported in thousands of MACs per pixel (kMACs/pixel).

As seen in Fig. 4 and 5, our architecture substantially outperforms every traditional codec across the entire bitrate range in both metrics, including VVC, the current international video compression standard. PACT approaches or surpasses ELIC-DCT performance in both Amplitude and Phase reconstruction, especially at low bitrates (below 1.0 BPP). This can be summarized via BD-Rates compared to a VVC baseline, where PACT achieves a 17.9% improvement in Amplitude SQNR and a 22.2% improvement in Phase MAE, compared to 21.6% and 20.9% for ELIC-DCT, respectively.

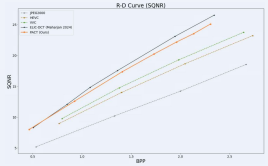


**Fig. 4.** BPP vs. SQNR

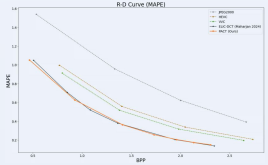


**Fig. 5.** BPP vs. MAPE

## Results, cont.

Detailed profiling of PACT and ELIC-DCT reveals a dramatic difference in model complexity, with PACT requiring 88% fewer trainable parameters and 96% fewer kMACs/pixel. This represents a state-of-the-art combination of compression performance and model efficiency.

| Method | BD-Rate (Amp SQNR) ↓ | BD-Rate (Phase MAE) ↓ | Parameters | Complexity (kMACs/pixel) ↓ | Enc. Complexity (kMACs/pixel) ↓ | Dec. Complexity (kMACs/pixel) ↓ |
|---|---|---|---|---|---|---|
| JPEG2000 | 52.0% | 69.9% | — | — | — | — |
| HEVC | 8.3% | 8.0% | — | — | — | — |
| VVC | 0.0% | 0.0% | — | — | — | — |
| ELIC-DCT | **-21.6%** | -20.9% | 33,005,378 | 469.64 | 139.26 | 261.82 |
| PACT (Ours) | -17.9% | **-22.2%** | **3,836,396** | **20.34** | **9.97** | **10.37** |

**Table 1.** Final results of PACT testing against ELIC-DCT and three traditional codecs. A BD-Rate of -17.9% corresponds to an average 17.9% bitrate savings compared to VVC for the same image reconstruction quality.

## Conclusions

These results demonstrate that learned SAR image compression can approach state-of-the-art reconstruction quality without sacrificing computational practicality. By integrating DCT preprocessing and an adaptive energy-guided architecture, PACT achieves competitive amplitude and phase fidelity relative to ELIC-DCT with 96% fewer operations, outperforming VVC and other classical codecs by a wide margin across all evaluated metrics.

Future work will extend PACT in two directions. First, the incorporation of cross-polarization side information, leveraging the complementary signal content across SAR polarization channels, is expected to improve reconstruction fidelity without proportional complexity increases. Second, further efficiency improvements targeting the entropy modeling and decoding pipeline will push PACT toward real-time deployment on resource-constrained platforms.

## References


[1] B. Bross, Y.-K. Wang, Y. Ye, S. Liu, J. Chen, G. J. Sullivan, and J.-R. Ohm, "Overview of the versatile video coding (VVC) standard and its applications," IEEE Transactions on Circuits and Systems for Video Technology, vol. 31, no. 10, pp. 3736–3764, 2021.

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## Other Information


This work was supported through a Graduate Research Assistantship by the Missouri Institute for Defense & Energy (MIDE).



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